Differences in antioxidant levels of fresh, frozen and freeze-dried strawberries and strawberry jam.

The present study was conducted to determine differences in antioxidant levels of fresh, frozen, and freeze-dried strawberries, and strawberry jam. Hydrophilic antioxidant activity (HAA) and lipophilic antioxidant activity (LAA) were measured using the ABTS/H■O■/HRP decoloration method. HAA and LAA were then summed to calculate the total antioxidant activity (TAA). Mean differences in HAA and LAA were analyzed using one-way analysis of variance and Dunnett's T3 pairwise comparisons. The mean TAA for freeze-dried strawberries based on an 'as consumed' weight (95% confidence interval [CI]: 29.58, 30.58) was significantly higher than for fresh (95% CI: 3.18, 3.66), frozen (95% CI: 2.58, 2.79), and jam (95% CI: 1.10, 1.22). The mean TAA based on dry weight for fresh strawberries (95% CI: 40.48, 46.67) was significantly higher than for freeze-dried (95% CI: 29.58, 30.58), frozen (95% CI: 24.62, 26.59), and jam (95% CI: 1.48, 1.64). Results agree with previous studies reporting that strawberries are a valuable source of antioxidants for consumers.